TARGET : JEE (M + A) : 2022

3 YEAR SRG ENTHUSIAST TEST SERIES

TEST TYPE: ONLINE | DATE : 19.12.2021 | PATTERN : JEE ADVANCED

Time : 3 Hours | Maximum Marks : 198

TEST - 14 (PAPER 1)

INSTRUCTIONS

READ THE INSTRUCTIONS CAREFULLY

QUESTION PAPER FORMAT AND MARKING SCHEME

1. The question paper has **three parts** : **PHYSICS**, **CHEMISTRY** and **MATHEMATICS**. Each part has **THREE SECTIONS**.
2. Carefully read the instructions given at the beginning of each section.
3. **Section I:** This section contains **SIX** questions. The answer to each question is a **ONLY ONE CORRECT OPTION.**

   Marking scheme : +3 for correct answer, 0 if not attempted and –1 in all other cases.
4. **Section II:** This section contains **SIX** questions. The Answer to each question is a **ONE OR MORE THAN ONE CORRECT OPTIONS**.

Marking scheme :	+4	for correct answer, 0 if not attempted and –2 in all other cases.
Partial Marks :	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks :	+2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks :	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
5. **Section III:** This section contains **SIX** questions. The answer to each question is a **NUMERICAL VALUE**.

   Marking scheme : +4 correct answer, 0 if not attempted and 0 in all other cases.

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SOME USEFUL CONSTANTS

Atomic No. H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17,Br = 35, Xe = 54, Ce = 58,

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23}\ JK^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8\ ms^{-1}$
- **Stefan–Boltzmann constant** $\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3}\ m\text{–}K$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34}\ J\text{–}s$

3 YEAR SRG ENTHUSIAST COURSE

TEST - 14 (PAPER 1) DATE : 19 - 12 - 2021

SYLLABUS

PHYSICS : Mathematical tools, Rectilinear and Projectile, Combine, Relative Motion, NLM and Friction, Work Energy and Power, Circular Motion, Center of Mass

CHEMISTRY : Full Syllabus

MATHS : Full Syllabus

PART-1 : PHYSICS

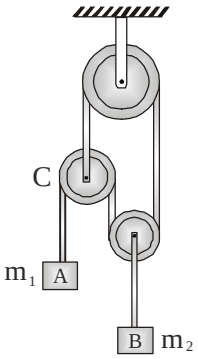
SECTION–I : (Maximum Marks : 18)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –1 In all other cases

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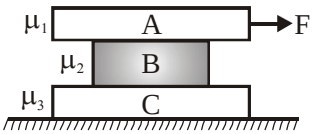
1. In the arrangement shown in figure neglect the masses of the pulley and string, and also friction. The accelerations of blocks A and B are



(A) g, g/2 (B) g/2, g

(C) 3g/2, 3g/4 (D) g, g

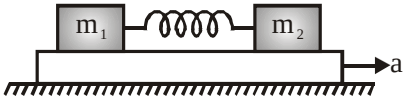
2. Given m_A = 30 kg, m_B = 10 kg, m_C = 20 kg. The coefficient of friction between A and B, μ_1 = 0.3, between B and C, μ_2 = 0.2 and between C and ground, μ_3 = 0.1. The least horizontal force F to start motion of any part of the system of three blocks resting upon one another as shown in figure is ($g = 10\ m/s^2$)



(A) 60 N (B) 90 N

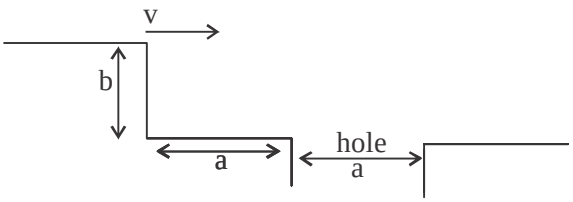
(C) 80 N (D) 150 N

3. Two blocks of masses m_1 and m_2 are connected with a massless spring and placed over a plank moving with an acceleration 'a' as shown in figure. The coefficient of friction between the blocks and platform is μ.



(A) Spring will be stretched if $a > \mu g$
(B) Spring will be compressed if $a \leq \mu g$
(C) Spring will neither be compressed nor be stretched for $a \leq \mu g$
(D) Spring will be in its natural length under all conditions

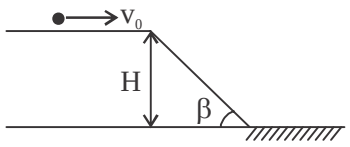
4. A particle is projected horizontally from a height b. The range of values of horizontal speed v so that it falls in the hole of length a is



(A) $\left(a\sqrt{\frac{g}{b}}, 2a\sqrt{\frac{g}{b}}\right)$ (B) $\left(a\sqrt{\frac{g}{2b}}, a\sqrt{\frac{2g}{b}}\right)$

(C) $\left(a\sqrt{\frac{g}{b}}, 3a\sqrt{\frac{g}{b}}\right)$ (D) $\left(b\sqrt{\frac{g}{b}}, a\sqrt{\frac{g}{b}}\right)$

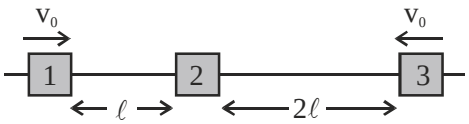
5. A block is in the form of a flat and inclined plane. A body travels with a speed v_0 as shown. The minimum value of angle β so that the particle does not hit the incline is



(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$

(C) $\tan^{-1}\frac{v_0}{\sqrt{2gH}}$ (D) $\tan^{-1}\left(\frac{1}{v_0}\sqrt{\frac{gH}{2}}\right)$

6. Three small bodies of identical masses can move along a straight line. The central body (2) is initially at rest and bodies 1 and 3 are at a distance l and 2l from the central body respectively. Bodies 1 and 3 move towards body 2 with speeds V_0 each. The collision between 1 and 2 is perfectly elastic and the collision between body 2 and 3 is perfectly inelastic. After all the collisions are over.



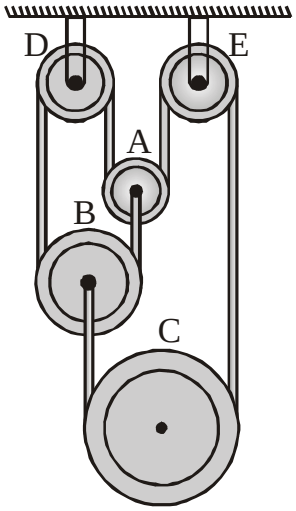
(A) All the bodies come to rest.
(B) The body 1 moves towards left, bodies 2 and 3 move towards right.
(C) Body 2 remains at rest and other bodies 1 and 3 turn back.
(D) All the bodies move towards right.

SECTION– II : (Maximum Marks : 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

7. In the pulley system shown in figure the movable pulleys A, B and C are of mass 1 kg each. D and E are fixed pulleys. The strings are light and inextensible. Choose the correct alternative(s). All pulleys are frictionless.



(A) Tension in the string is 6.5 N

(B) Acceleration of pulley A is g/3 downward

(C) Acceleration of pulley B is g/6 upward

(D) Acceleration of pulley C is g/3 upward

8. The Fig. shows a string of equally placed beads of mass m, separated by distance d. The beads are free to slide without friction on a thin wire. A constant force F acts on the first bead initially at rest till it makes collision with the second bead. The second bead then collides with the third and so on. Suppose that all collisions are elastic,



(A) Speed of the first bead immediately before and immediately after its collision with the second bead is $\sqrt{\frac{2Fd}{m}}$ and zero respectively.

(B) Speed of the first bead immediately before and immediately after its collision with the second bead is $\sqrt{\frac{2Fd}{m}}$ and $\frac{1}{2}\sqrt{\frac{2Fd}{m}}$ respectively.

(C) Speed of the second bead immediately after its collision with third bead is zero.

(D) The average speed of the first bead is $\frac{1}{2}\sqrt{\frac{2Fd}{m}}$.

9. A set of n–identical cubical blocks lies at rest parallel to each other along a line on a smooth horizontal surface. The separation between the near surfaces of any two adjacent blocks is L. The block at one end is given a speed v towards the next one at time t = 0. All collisions are completely inelastic, then

(A) The last block starts moving at $t = n\,(n-1)\frac{L}{2v}$

(B) The last block starts moving at $t = (n-1)\frac{L}{v}$

(C) The centre of mass of the system will have a final speed $\frac{v}{n}$

(D) The centre of mass of the system will have a final speed v.

10. In a one dimensional collision between two identical particles A and B, B is stationary and A has momentum p before impact. During impact, B gives impulse J to A.

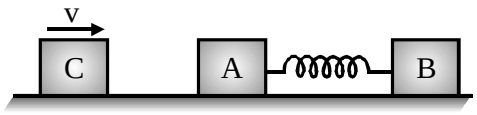
(A) The total momentum of the 'A plus B' system is p before and after the impact, and (p–1) during the impact

(B) During the impact A gives impulse J to B

(C) The coefficient of restitution is $\frac{2J}{p}-1$

(D) The coefficient of restitution is $\frac{J}{p}+1$

11. Two blocks A and B each of mass m, are connected by a massless spring of natural length L and spring constant k. The blocks are initially resting on a smooth horizontal floor with the spring at its natural length, as shown in fig. A third identical block C, also of mass m, moves on the floor with a speed v along the line joining A and B, and collides elastically with A. Then :–



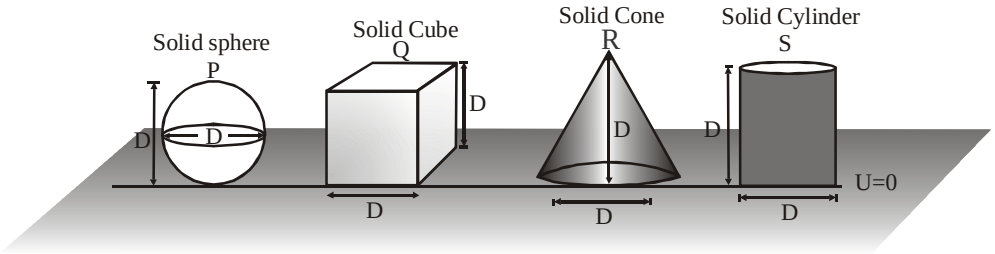
(A) The kinetic energy of the A–B system, at maximum compression of the spring, is zero

(B) The kinetic energy of the A–B system, at maximum compression of the spring, is $\frac{mv^2}{4}$

(C) The maximum compression of the spring is $v\sqrt{\left(\frac{m}{k}\right)}$

(D) The maximum compression of the spring is $v\sqrt{\frac{m}{2k}}$

12. Assuming potential energy 'U' at ground level to be zero.



All objects are made up of same material.

U_P = Potential energy of solid sphere

U_Q = Potential energy of solid cube

U_R = Potential energy of solid cone

U_S = Potential energy of solid cylinder

(A) $U_S > U_P$

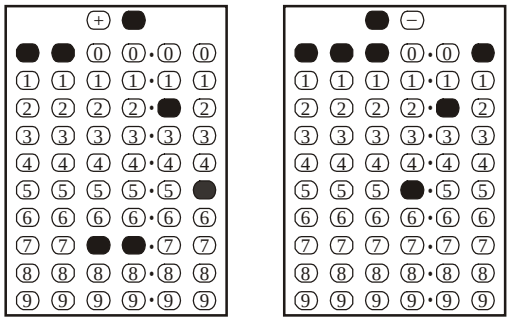
(B) $U_Q > U_S$

(C) $U_P > U_Q$

(D) $U_S > U_R$

SECTION-III : (Maximum Marks: 24)

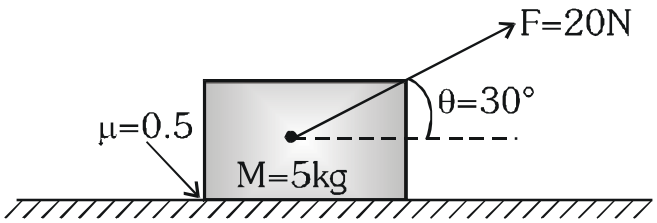
- This section contains **SIX (06)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.
  For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



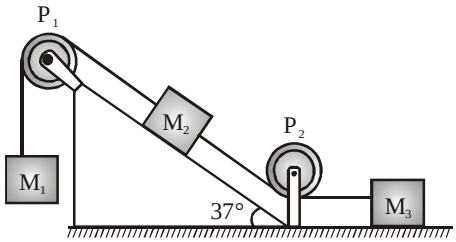
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +4 If ONLY the correct numerical value is entered as answer.
  Zero Marks : 0 In all other cases.

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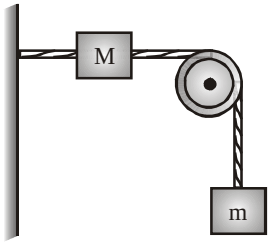
13. Calculate the force of friction (in Newton) for shown situation.



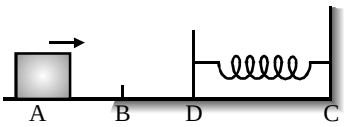
14. Masses M_1, M_2 and M_3 are connected by strings of negligible mass which pass over massless and frictionless pulleys P_1 and P_2 as shown in fig. The masses move such that the portion of the string between P_1 and P_2 is parallel to the inclined plane and the portion of the string between P_2 and M_3 is horizontal. The masses M_2 and M_3 are 4.0 kg each and the coefficient of kinetic friction between the masses and the surfaces is 0.25. The inclined plane makes an angle of 37° with the horizontal. If the mass M_1 moves downwards with a uniform velocity, find the mass of M_1. (in kg)



15. A string, with one end fixed on a rigid wall, passing over a fixed frictionless pulley at a distance of 2m from the wall, has a point mass M = 2 kg attached to it at a distance of 1m from the wall. A mass m = 0.5 kg attached at the free end is held at rest so that the string is horizontal between the wall and the pulley and vertical beyond the pulley. What will be the speed (in m/s) with which the mass M will hit the wall when the mass m is released ?

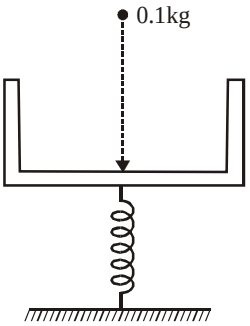


16. A 0.5 kg block slides from the point A (see figure) on a horizontal track with an initial speed of 3 m/s towards a weightless horizontal spring of length 1 m and force constant 2 N/m. The part AB of the track is frictionless and the part BC has the coefficients of static and kinetic friction as 0.22 and 0.2 respectively. If the distance AB and BD are 2m and 2.14 m respectively, find the total distance (in meter) through which the block moves before it comes to rest completely. (Take g = 10 m/s^2).



17. A ball of mass 100 g is projected vertically upwards from the ground with a velocity of 49m/s. At the same time another identical ball is dropped from a height of 98 m to fall freely along the same path as that followed by the first ball. After some time the two balls collide and stick together and finally fall to the ground. Find the time of flight (in second) of the masses.

18. A massless platform is kept on a light elastic spring, shown in Fig. When a sand particle of 0.1 kg mass is dropped on the pan from a height 0.24 m, the particle strikes the pan, and the spring compresses by 0.01 m. From what height (in meter) should the particle be dropped to cause a compression of 0.04 m ?



PART-2 : CHEMISTRY

SECTION–I : (Maximum Marks : 18)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- Answer to each question will be evaluated according to the following marking scheme :

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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –1 In all other cases

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1. A mixture containing 0.05 mole of $K_2Cr_2O_7$ and 0.02 mole of $KMnO_4$ was treated with excess of KI in acidic medium. The liberated iodine required 1.0 L of $Na_2S_2O_3$ solution for titration. Concentration of $Na_2S_2O_3$ solution was :

(A) 0.40 mol L^{-1} (B) 0.20 mol L^{-1}

(C) 0.25 mol L^{-1} (D) 0.30 mol L^{-1}

2. The energy of separation of an electron in a hydrogen like atom in excited state is 3.4 eV. The de-Broglie wave length (in Å) associated with the electron is :

(Given radius of first orbit of H-atom is 0.53Å)

(A) 3.33 (B) 6.66

(C) 13.31 (D) None of these

3. For a real gas (mol mass = 60) if density at critical point is $0.80\ g/cm^3$ and its $T_c = \frac{4\times10^5}{821}K$, then van der Waal's constant a (in atm L^2mol^{-2}) is

(A) 0.3375 (B) 3.375

(C) 1.68 (D) 0.025

4. What volume of 75% alcohol by weight $(d = 0.80\,g/cm^3)$ must be used to prepare $150cm^3$ of 30% alcohol by mass $(d = 0.90\,g/cm^3)$?

(A) 67.5 mL (B) 56.25 mL

(C) 44.44 mL (D) 32.2 mL

5. Using MO theory predict which of the following species has the shortest bond length ?

(A) O_2^{2+} (B) O_2^{+}

(C) O_2^{-} (D) O_2^{2-}

6. The bond dissociation energy of $B-F$ in BF_3 is $646\,kJmol^{-1}$ where as that of $C-F$ in CF_4 is 515 kJ mol^{-1}. The correct reason for higher $B-F$ bond dissociation energy as compared to that of $C-F$ is :

(A) Smaller size of B atom as compared to that of C atom

(B) Stronger σ bond between B and F in BF_3 as compared to that between C and F in CF_4

(C) Significant $p\pi-p\pi$ interaction between B and F in BF_3 whereas there is no possibility of such interaction between C and F in CF_4

(D) Lower degree of $p\pi-p\pi$ interaction between B and F in BF_3 than that between C and F in CF_4

SECTION– II : (Maximum Marks : 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

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Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

7. Which of the following statement is / are correct ?

(A) Tl^{3+} salts are oxidising agents (B) Ga^{+} salts are reducing agents

(C) Pb^{4+} salts are better oxidising agents (D) As^{5+} salts are oxidising agents

8. Which of the following oxides are amphoteric ?

(A) BeO (B) SnO

(C) ZnO (D) Al_2O_3

9. Molecules with see-saw shape are :

(A) $XeOF_4$ (B) SF_4

(C) $XeOF_2$ (D) XeO_2F_2

10. In which of the following species bond angle is expected to be more than 120° ?

(A) PCl_4^+ (B) NO_2^-

(C) NO_2^+ (D) XeF_2

11. Identify correct order of bond angles

(A) $Cl_2O > F_2O$ and $F_2O < H_2O$

(B) $AsI_3 > AsBr_3 > AsCl_3$

(C) $NO_2^+ > NO_2^-$

(D) $H_b\hat{B}H_b > H_t\hat{B}H_t$; where H_t is terminal Hydrogen of B_2H_6 and H_b is the bridging Hydrogen of B_2H_6

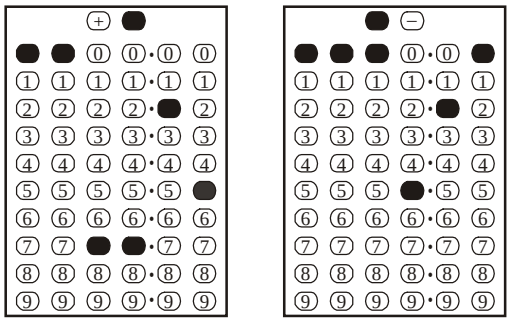
12. Which among the following compounds are paramagnetic ?

(A) KO_2 (B) K_2O_2

(C) K_2O (D) NO_2

SECTION-III : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.
  For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +4 If ONLY the correct numerical value is entered as answer.
  Zero Marks : 0 In all other cases.

13. 2 mole, equimolar mixture of $Na_2C_2O_4$ and $H_2C_2O_4$ required V_1L of 0.1 M $KMnO_4$ in acidic medium for complete oxidation. The same amount of the mixture required V_2L of 0.2 M NaOH for neutralization. Then the value of $(V_1 + V_2)$ is

14. If I excitation energy for the H-like (hypothetical) sample is 24 eV, then binding energy in III excited state in
eV is :

15. The van der Waal's constant 'b' of a gas is $4\pi \times 10^{-4}$ L / mol the centres of the two molecules approach each other is 10^{-x} m . What is x ? $N_A = 6 \times 10^{23}$

16. The wave function of atomic orbital of H like atoms is given as under

$$\psi_{2s} = \frac{1}{4\sqrt{2\pi}} z^{3/2} [2 - Zr] e^{Zr/2}$$

Given that the radius is in Å, then radius of nodal surface for He^+ ion is (in Å)

17. The first ionisation energy of Li is 5.41 eV and electron affinity of chlorine is 3.61 eV.

The ΔH in $kJmol^{-1}$ for the reaction :

$Li(g) + Cl(g) \rightarrow Li^+(g) + Cl^-(g)$ is 1.737×10^x kJ mol^{-1}

What is the value of x ?

18. Among the triatomic molecules / ions , $BeCl_2, N_3^-, N_2O, NO_2^+, O_3, SCl_2, ICl_2^-, I_3^-$ and XeF_2, the total number of linear molecule (s) / ion(s) where the hybridization of the central atom does not have contribution from the
d-orbital(s) is :

PART-3 : MATHEMATICS

SECTION–I : (Maximum Marks : 18)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –1 In all other cases

---

1. If terms independent of x in the expansion of $\left(3x-\frac{1}{x}\right)^{20}$ and $\left(x+\frac{\sqrt[9]{3^{10}}}{x}\right)^{18}$ are A and B respectively then $\left(\frac{9}{38}A+B\right)$ equals

(A) $3^{10}\cdot{}^{19}C_8$ (B) $3^{10}\cdot{}^{19}C_9$

(C) $3^{9}\cdot{}^{20}C_8$ (D) $3^{9}\cdot{}^{19}C_{10}$

2. There are 3 women participating in a chess tournament. Every participant played 3 games with other participants. Number of games that the men played among themselves proved to exceed by 27 number of games that men played with women. Then number of men playing in the tournament is _____

(A) 3 (B) 5

(C) 9 (D) 12

3. The least integer 'a' for which $1+\log_3\left(x^2+1\right)\le\log_3\left(ax^2+4x+a\right)$ is true for all real values of x is

(A) 3 (B) 4

(C) 5 (D) 6

4. The least value of x satisfying $\frac{1}{[x]}+\frac{1}{[2x]}=\{x\}+\frac{1}{3}$ is N, then number of positive divisors of 24N is

(where $[\cdot]$ is the greatest integer function and $\{x\}$ is fractional part of x)

(A) 2 (B) 4

(C) 6 (D) 8

5. If $S_n(\theta)=\sum_{r=1}^{n}\frac{\cos(r-1)\theta}{(\cos\theta)^{r-1}}$, then sum of the series $\left[S_3\left(\frac{\pi}{3}\right)\right]+\left[S_9\left(\frac{\pi}{3}\right)\right]+\left[S_{15}\left(\frac{\pi}{3}\right)\right]+\ldots+\left[S_{2025}\left(\frac{\pi}{3}\right)\right]$ is

(where $[\cdot]$ denotes the greatest integer function)

(A) 0 (B) 1

(C) 3 (D) 2025

6. If $2\cos^4 x+\sqrt{5}\left(\sqrt{5}-1\right)=\cos^2 x\left(\sqrt{5}+1+2\sin x\right)+\left(\sqrt{5}-1\right)\sin x$, then

$\sin^{10}x+4\sin^9 x+6\sin^8 x+4\sin^7 x+\sin^6 x+\sin x+1$ equals

(A) 1 (B) 2

(C) 3 (D) 4

SECTION– II : (Maximum Marks : 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
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Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

7. In a game a person P at (x, y) can jump to any one of the following points relatively $(x + 2, y)$, $(x + 4, y)$, $(x, y + 2)$, $(x, y + 4)$. If P is at (1, 1) then the number of ways to reach the point.

(A) (5, 5) is 14 (B) (5, 5) is 24

(C) (7, 8) is 0 (D) (9, 9) through (5, 5) is 196

8. If the equation $\left(2+\sin^2 x\right)^2-(a-3)\left(2+\sin^2 x\right)\left(1+\sin^2 x\right)+(a-4)\left(1+\sin^2 x\right)^2=0$ has atleast one real root then the complete set of values of a is $[\alpha, \beta]$ then

(A) $\alpha+\beta=11$ (B) $\beta-\alpha=\frac{1}{2}$

(C) $\alpha+\beta=\frac{33}{2}$ (D) $\beta-\alpha=1$

9. Let $S=\sum_{k=1001}^{3001}\left(\frac{1}{k}\right)$, then

(A) $S > 1$ (B) $S < 1$

(C) $S>\frac{4}{3}$ (D) $S<\frac{4}{3}$

10. Let $N=2^2\times3^3\times7^4$, then

(A) Number of factors of N^2 which are less than N and are not the factors of N is 98

(B) Number of factors of N which are of the form $4K + 1$, $K \in N$ is 9

(C) Number of factors of N which are multiples of 54 is 10

(D) Number of factors of N which are of the form $4K + 1$, $K \in N$ is 8

11. The series $\frac{8}{5}+\frac{16}{65}+\frac{24}{325}+.....\frac{8n}{4n^4+1}$

(A) Sum of infinite number of terms of series is 4

(B) Sum of n terms of series is less than 2 $\forall\ n \in N$

(C) Sum to n terms of series cannot be an integer for any $n \in N$

(D) Sum of infinite number of terms of series is less than 3

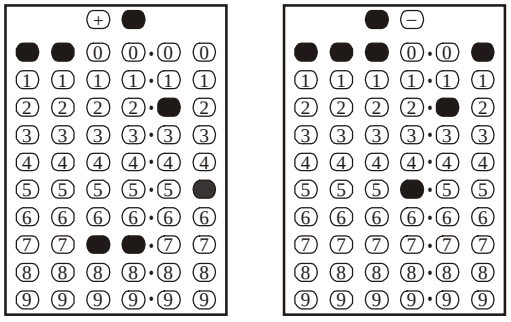
12. Consider 5 points in plane are situated so that no two of the straight lines joining them are parallel, perpendicular, or co incident. From each point perpendiculars the drawn to all the lines joining the other four points. If k is the maximum number of intersections that these perpendiculars can have ? Then

(A) More than 310 (B) k is divisible by 5

(C) k is less than 400 (D) k is even number

SECTION-III : (Maximum Marks: 24)

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13. If $S=\lim\limits_{n\to\infty}\sum\limits_{r=2}^{n}\frac{{}^{r}C_2}{3^r}$. Then value of 48 S is

14. If α, β are two roots of the equation $1!+2!+3!+.....(x-1)!+x!=k^2$, $k\in I^+$. Where $\alpha<\beta$ and if α_1, α_2, α_3, α_4 are roots of equation $\left(a+\sqrt{b}\right)^{x^2-\left(\alpha+3\alpha+5\alpha^2\right)}+\left(a-\sqrt{b}\right)^{x^2-3\beta}=2a$, where $a^2-b=1$.
Then the value of $|\alpha_1+\alpha_2+\alpha_2+\alpha_3+\alpha_1\alpha_2\alpha_3\alpha_4|$ is equal to

15. The real value of x satisfying $\sqrt[3]{20x+\sqrt[3]{20x+13}}=13$ can be expressed as $\frac{a}{b}$ where a and b are relatively prime positive integers. Then the value of $(a-100b)$ is

16. Number of order pairs (x, y) which satisfy the equation $x^2+2x\sin(xy)+1=0$, where $y\in[0,\ 2\pi]$ is

17. Let $P(x)=x^6-x^5-x^3-x^2-x$ and α, β, γ, δ are the roots of the equation $x^4-x^3-x^2-1=0$, then $P(\alpha)+P(\beta)+P(\gamma)+P(\delta)=$

18. The result of 22 hockey matches are to be predicted. Given that result can be win, lose or draw. If p is number of different forecast which contain exactly 18 correct results and q is number of different forecast which contains exactly 19 correct results. Then value of $\left(\frac{4p}{q}\right)$ is

3 YEAR SRG ENTHUSIAST COURSE

TEST - 14 (PAPER 1)	DATE : 19 - 12 - 2021

SYLLABUS

PHYSICS	**:**	Mathematical tools, Rectilinear and Projectile, Combine, Relative Motion, NLM and Friction, Work Energy and Power, Circular Motion, Center of Mass
CHEMISTRY	**:**	Full Syllabus
MATHS	**:**	Full Syllabus

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

TARGET : JEE (M + A) : 2022

3 YEAR SRG ENTHUSIAST TEST SERIES

TEST TYPE: ONLINE | DATE : 19.12.2021 | PATTERN : JEE ADVANCED

Time : 3 Hours | Maximum Marks : 198

TEST - 14 (PAPER 1)

ANSWER

PART 1 - PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	D	A	D	B	D	A
SECTION-II	Q.	7	8	9	10	11	12
	A.	ABD	AC	AC	BC	BD	ABD
SECTION-III	Q.	13	14	15	16	17	18
	A.	17.3	4.2	3.4	4.24	6.53	3.96

PART 2 - CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6
	A.	A	B	B	A	A	C
SECTION-II	Q.	7	8	9	10	11	12
	A.	ABC	ABCD	BD	CD	ABC	AD
SECTION-III	Q.	13	14	15	16	17	18
	A.	18.00	12.00	10	1.0	2	4

PART 3 - MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	B	C	C	B	A	B
SECTION-II	Q.	7	8	9	10	11	12
	A.	ACD	BC	AD	ABC	BCD	ABC
SECTION-III	Q.	13	14	15	16	17	18
	A.	18	80	46	02	06	38

PART-1 : PHYSICS

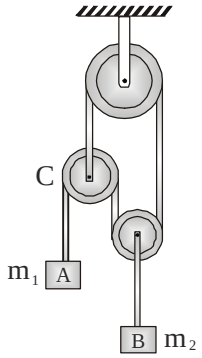
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Negative Marks : –1 In all other cases

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1. In the arrangement shown in figure neglect the masses of the pulley and string, and also friction. The accelerations of blocks A and B are

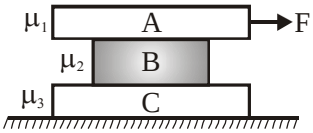


(A) g, g/2 (B) g/2, g

(C) 3g/2, 3g/4 (D) g, g

Ans. D

2. Given m_A = 30 kg, m_B = 10 kg, m_C = 20 kg. The coefficient of friction between A and B, μ_1 = 0.3, between B and C, μ_2 = 0.2 and between C and ground, μ_3 = 0.1. The least horizontal force F to start motion of any part of the system of three blocks resting upon one another as shown in figure is ($g = 10\ m/s^2$)

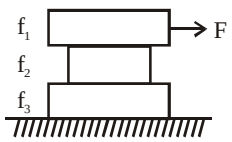


(A) 60 N (B) 90 N

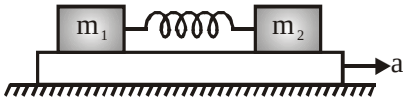
(C) 80 N (D) 150 N

Ans. A

Sol. Maximum value of $f_1 = 0.3 \times 30 \times 10 = 90$ N
Maximum value of $f_2 = 0.2 \times 40 \times 10 = 80$ N
Maximum value of $f_3 = 0.1 \times 60 \times 10 = 60$ N
$\Rightarrow$ Least horizontal force F to start motion = 60N



3. Two blocks of masses m_1 and m_2 are connected with a massless spring and placed over a plank moving with an acceleration 'a' as shown in figure. The coefficient of friction between the blocks and platform is μ.

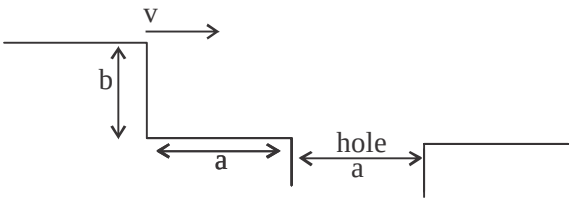


(A) Spring will be stretched if $a > \mu g$
(B) Spring will be compressed if $a \leq \mu g$
(C) Spring will neither be compressed nor be stretched for $a \leq \mu g$
(D) Spring will be in its natural length under all conditions

Ans. D

Sol. Let a £ mg (i.e. friction is static) then both the blocks are at rest w.r.t. plank. Therefore spring will be in its natural length.
Now let a > mg (i.e. friction is kinetic) then both the blocks are moving with same acceleration w.r.t. plank. In this case spring force is equal to zero.

4. A particle is projected horizontally from a height b. The range of values of horizontal speed v so that it falls in the hole of length a is



(A) $\left(a\sqrt{\frac{g}{b}}, 2a\sqrt{\frac{g}{b}}\right)$ (B) $\left(a\sqrt{\frac{g}{2b}}, a\sqrt{\frac{2g}{b}}\right)$

(C) $\left(a\sqrt{\frac{g}{b}}, 3a\sqrt{\frac{g}{b}}\right)$ (D) $\left(b\sqrt{\frac{g}{b}}, a\sqrt{\frac{g}{b}}\right)$

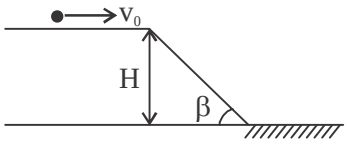
Ans. B

Sol. $t = \sqrt{\frac{2b}{g}}$

$$a < vt < 2a \Rightarrow a < v\sqrt{\frac{2b}{g}} < 2a$$

$$v \in \left(a\sqrt{\frac{g}{2b}}, a\sqrt{\frac{2g}{b}}\right)$$

5. A block is in the form of a flat and inclined plane. A body travels with a speed v_0 as shown. The minimum value of angle β so that the particle does not hit the incline is



(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$

(C) $\tan^{-1}\frac{v_0}{\sqrt{2gH}}$ (D) $\tan^{-1}\left(\frac{1}{v_0}\sqrt{\frac{gH}{2}}\right)$

Ans. D

Sol. If it does not hit the incline, it travels a vertical distance of H, so that the time of flight is

$$t=\sqrt{\frac{2H}{g}}$$

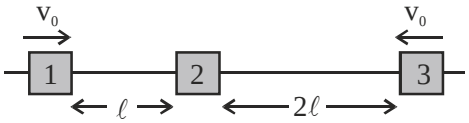
The initial velocity is horizontal, so the horizontal range is $=v_0t=v_0\sqrt{\frac{2H}{g}}$

The minimum value of range so that it does not hit the incline is $\frac{H}{\tan\beta}$

$$\Rightarrow \tan\beta=\frac{H}{v_0\sqrt{\frac{2H}{g}}}=\frac{1}{v_0}\sqrt{\frac{gH}{2}}$$

$$\beta=\tan^{-1}\left(\frac{1}{v_0}\sqrt{\frac{gH}{2}}\right)$$

6. Three small bodies of identical masses can move along a straight line. The central body (2) is initially at rest and bodies 1 and 3 are at a distance l and 2l from the central body respectively. Bodies 1 and 3 move towards body 2 with speeds V_0 each. The collision between 1 and 2 is perfectly elastic and the collision between body 2 and 3 is perfectly inelastic. After all the collisions are over.



(A) All the bodies come to rest.
(B) The body 1 moves towards left, bodies 2 and 3 move towards right.
(C) Body 2 remains at rest and other bodies 1 and 3 turn back.
(D) All the bodies move towards right.

Ans: A

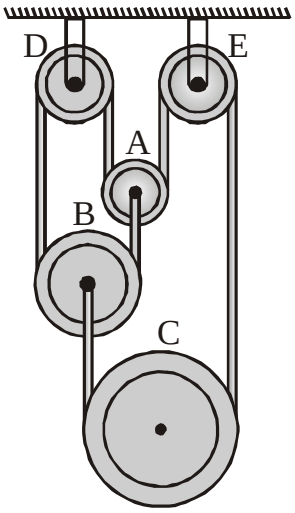
Sol: After the collision between 1 and 2, mass 1 stops and mass 2 proceeds towards right with speed v_0 to collide with mass 3. The collision between mass 2 and 3 is perfectly inelastic. Hence 2 and 3 also come to rest after the collision.

SECTION– II : (Maximum Marks : 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

7. In the pulley system shown in figure the movable pulleys A, B and C are of mass 1 kg each. D and E are fixed pulleys. The strings are light and inextensible. Choose the correct alternative(s). All pulleys are frictionless.



(A) Tension in the string is 6.5 N

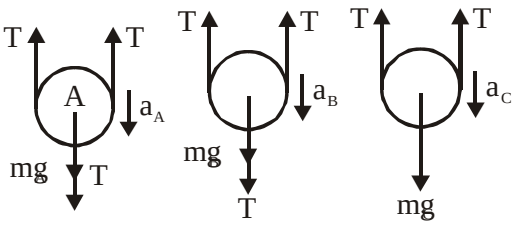
(B) Acceleration of pulley A is g/3 downward

(C) Acceleration of pulley B is g/6 upward

(D) Acceleration of pulley C is g/3 upward

Ans. ABD

Sol. Let tension in string be T



Here $(T)a_A + (T)a_B + (2T)a_C = 0 \Rightarrow a_A + a_B + 2a_C = 0$

$m_Ag - T = m_Aa_A$, $m_Bg - T = m_Ba_B$, $m_Cg - 2T = m_C a_C$

$\Rightarrow$ T = 6.5 N, $a_A = \frac{g}{3}$, $a_B = \frac{g}{3}$, $a_C = -\frac{g}{3}$

8. The Fig. shows a string of equally placed beads of mass m, separated by distance d. The beads are free to slide without friction on a thin wire. A constant force F acts on the first bead initially at rest till it makes collision with the second bead. The second bead then collides with the third and so on. Suppose that all collisions are elastic,



(A) Speed of the first bead immediately before and immediately after its collision with the second bead is $\sqrt{\frac{2Fd}{m}}$ and zero respectively.

(B) Speed of the first bead immediately before and immediately after its collision with the second bead is $\sqrt{\frac{2Fd}{m}}$ and $\frac{1}{2}\sqrt{\frac{2Fd}{m}}$ respectively.

(C) Speed of the second bead immediately after its collision with third bead is zero.

(D) The average speed of the first bead is $\frac{1}{2}\sqrt{\frac{2Fd}{m}}$.

Ans. AC

Sol. In elastic head on collision if the masses of the colliding bodies are equal, the velocities after collision are interchanged.

For Ist bead, $Fd = \frac{1}{2}mu^2 \Rightarrow u = \sqrt{\frac{2Fd}{m}}$

9. A set of n–identical cubical blocks lies at rest parallel to each other along a line on a smooth horizontal surface. The separation between the near surfaces of any two adjacent blocks is L. The block at one end is given a speed v towards the next one at time t = 0. All collisions are completely inelastic, then

(A) The last block starts moving at $t = n\,(n-1)\frac{L}{2v}$

(B) The last block starts moving at $t = (n-1)\frac{L}{v}$

(C) The centre of mass of the system will have a final speed $\frac{v}{n}$

(D) The centre of mass of the system will have a final speed v.

Ans. AC

Sol. $t_1 = \frac{L}{v}$ (time for Ist collision)

$t_2 = \frac{2L}{v}$ (time for IInd collision)

$t_3 = \frac{3L}{v}$ (time for 3rd collision)

$t_{(n-1)} = \frac{L}{v}(n-1)$ (time for (n^{th}) collision)

$\sum_{i-1}^{h} t_i = \frac{n(n-1)}{2}\frac{L}{v}$

10. In a one dimensional collision between two identical particles A and B, B is stationary and A has momentum p before impact. During impact, B gives impulse J to A.

(A) The total momentum of the 'A plus B' system is p before and after the impact, and (p–1) during the impact.

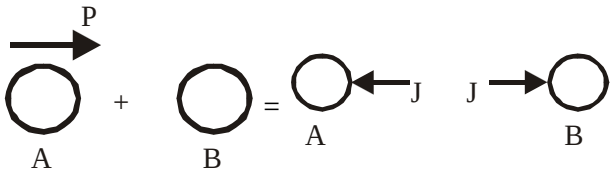
(B) During the impact A gives impulse J to B

(C) The coefficient of restitution is $\frac{2J}{p}-1$

(D) The coefficient of restitution is $\frac{J}{p}+1$

Ans. BC

Sol.

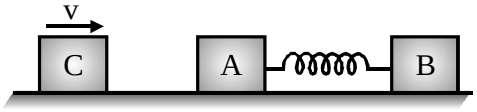


For A : $P - J = mv_1$ ....(i)

for B : $J = mv_2$ ....(ii)

$$\therefore e = -\left(\frac{v_2 - v_1}{u_2 - u_1}\right) = -\left[\frac{\frac{J}{m} - \left(\frac{P-J}{m}\right)}{0 - \frac{P}{m}}\right] = \frac{2J}{p} - 1$$

11. Two blocks A and B each of mass m, are connected by a massless spring of natural length L and spring constant k. The blocks are initially resting on a smooth horizontal floor with the spring at its natural length, as shown in fig. A third identical block C, also of mass m, moves on the floor with a speed v along the line joining A and B, and collides elastically with A. Then :–



(A) the kinetic energy of the A–B system, at maximum compression of the spring, is zero

(B) the kinetic energy of the A–B system, at maximum compression of the spring, is $\frac{mv^2}{4}$

(C) the maximum compression of the spring is $v\sqrt{\left(\frac{m}{k}\right)}$

(D) the maximum compression of the spring is $v\sqrt{\frac{m}{2k}}$

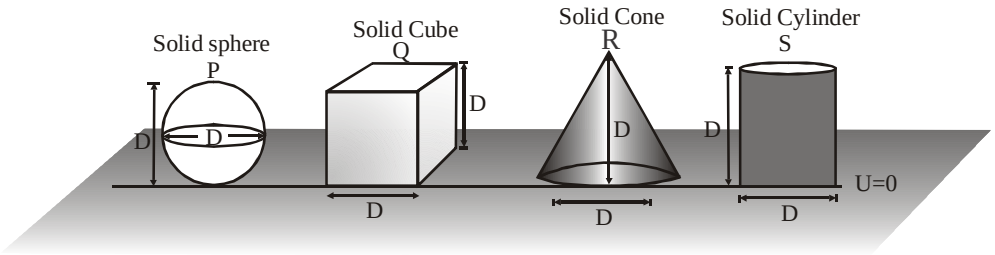
Ans. BD

Sol. At the time of maximum compression,

COLM : $mu = 2mv$ (for A & B) $\Rightarrow$ $v = u/2$

COME : $\frac{1}{2}mu^2 = \frac{1}{2}2mv^2 + \frac{1}{2}kx^2 \Rightarrow x = v\sqrt{\frac{m}{2k}}$

12. Assuming potential energy 'U' at ground level to be zero.



All objects are made up of same material.

U_P = Potential energy of solid sphere

U_Q = Potential energy of solid cube

U_R = Potential energy of solid cone

U_S = Potential energy of solid cylinder

(A) $U_S > U_P$

(B) $U_Q > U_S$

(C) $U_P > U_Q$

(D) $U_S > U_R$

Ans. ABD

Sol. PE of solid sphere = $mgR = m_p g\frac{D}{2} = rgD^4\left(\frac{\pi}{12}\right)$

PE of solid cube = $m_Q g\frac{D}{2} = rgD^4\left(\frac{1}{2}\right)$

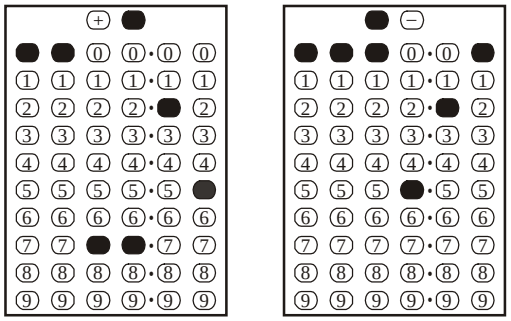
PE of solid cone = $m_R g\frac{D}{4} = rgD^4\left(\frac{\pi}{48}\right)$

PE of solid cylinder = $m_S g\frac{D}{2} = rgD^4\left(\frac{\pi}{8}\right)$

SECTION-III : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

  For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.

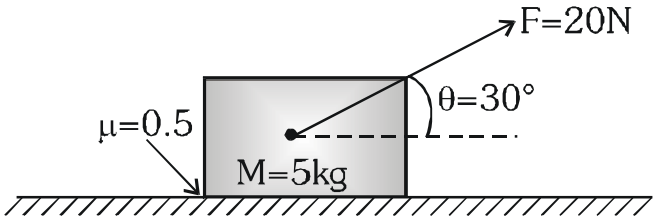


- Answer to each question will be evaluated according to the following marking scheme:

  Full Marks : +4 If ONLY the correct numerical value is entered as answer.

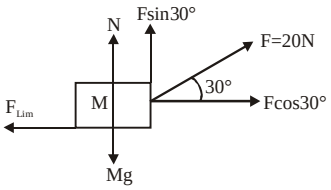
  Zero Marks : 0 In all other cases.

13. Calculate the force of friction (in Newton) for shown situation.



Ans. 17.3

Sol.



$F_{lim} = mN = m(Mg - Fsin30°) = 0.5(5 \times 9.8 - 20 \frac{1}{2})$

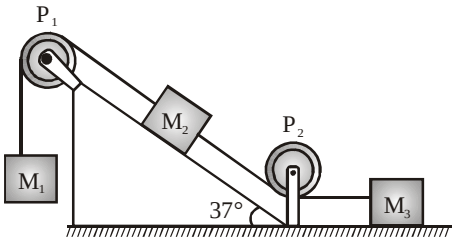
$= 0.5\ (49.0 - 10) = 0.5\ (39) = 19.5$ N

$F_{applied} = F \cos 30° = \frac{20\sqrt{3}}{2} = 17.3$ N

Since $F_{applied} < F_{lim}$

$\therefore$ Force of friction = $F_{applied} = 17.3$ N

14. Masses M_1, M_2 and M_3 are connected by strings of negligible mass which pass over massless and frictionless pulleys P_1 and P_2 as shown in fig. The masses move such that the portion of the string between P_1 and P_2 is parallel to the inclined plane and the portion of the string between P_2 and M_3 is horizontal. The masses M_2 and M_3 are 4.0 kg each and the coefficient of kinetic friction between the masses and the surfaces is 0.25. The inclined plane makes an angle of 37° with the horizontal. If the mass M_1 moves downwards with a uniform velocity, find the mass of M_1. (in kg)



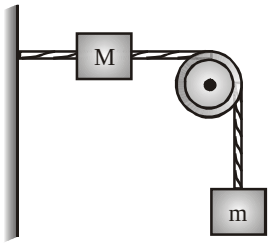
Ans. 4.2

Sol. $M_1 g = \mu M_2 g \cos\theta + \mu M_3 g + M_2 g \sin\theta$

$\Rightarrow M_1 = \mu (M_2 \cos\theta + M_3) + M_2 \sin\theta$

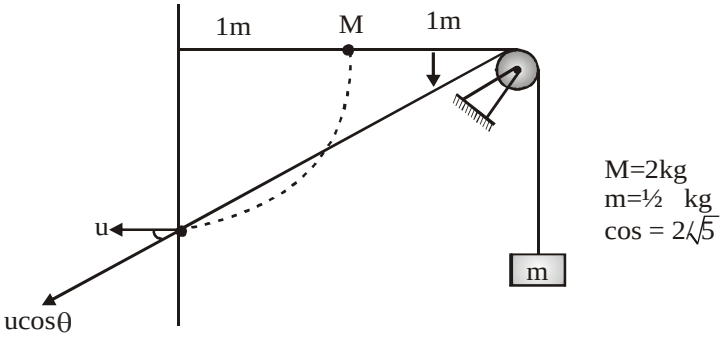
$= 0.25 (4 \times \frac{4}{5} + 4) + 4 \times \frac{3}{5} = \frac{21}{5} = 4.2$ kg

15. A string, with one end fixed on a rigid wall, passing over a fixed frictionless pulley at a distance of 2m from the wall, has a point mass M = 2kg attached to it at a distance of 1m from the wall. A mass m = 0.5 kg attached at the free end is held at rest so that the string is horizontal between the wall and the pulley and vertical beyond the pulley. What will be the speed (in m/s) with which the mass M will hit the wall when the mass m is released ?



Ans. 3.4

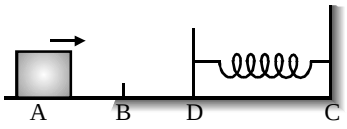
Sol. COME : $K_i + U_i = K_f + U_f$



$\Rightarrow \quad 0 + Mg \times R_1 = 0 + mg\left(\sqrt{5}-1\right) + \frac{1}{2}Mu^2 + \frac{1}{2}m(u\cos\theta)^2$

$\Rightarrow \quad u = 3.4$ m/s

16. A 0.5 kg block slides from the point A (see figure) on a horizontal track with an initial speed of 3 m/s towards a weightless horizontal spring of length 1 m and force constant 2 N/m. The part AB of the track is frictionless and the part BC has the coefficients of static and kinetic friction as 0.22 and 0.2 respectively. If the distance AB and BD are 2m and 2.14 m respectively, find the total distance (in meter) through which the block moves before it comes to rest completely. (Take g = 10 m/s^2).



Ans. 4.24

Sol. WET : $W_N + W_{Mg} + W_f + W_{sp} = DKE$

$0 + 0 - m_k.mg\,(2.14 + x) + 0 - \frac{1}{2}kx^2 = 0 - \frac{1}{2}mv^2$

$\Rightarrow \quad x = 0.1$ m

At x = 1m, $F_{spring} = kx = 2 \times 0.1 = 0.2$ N

$F_{S.F.} = m_s.mg = 0.22 \times \frac{1}{2} \times 10 = 1.1$ N

Hence the block stops after compressing the spring.

$\therefore$ Total distance travelled by block when it stops

$= 2 + 2.14 + 0.1 = 4.24$ m

17. A ball of mass 100 g is projected vertically upwards from the ground with a velocity of 49m/s. At the same time another identical ball is dropped from a height of 98 m to fall freely along the same path as that followed by the first ball. After some time the two balls collide and stick together and finally fall to the ground. Find the time of flight (in second) of the masses.

Ans. 6.53

Sol. $\left(\frac{1}{2}gt^2\right)+\left(49t-\frac{1}{2}gt^2\right)=98$

$\Rightarrow \quad t = 2s$

$\vec{u}_1 = gt(\hat{j}) = -19.6\hat{j}\,m/s$

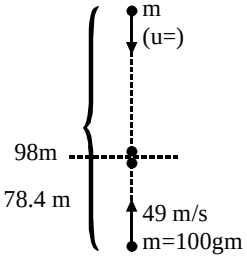
$\vec{u}_2 = (u - gt)\hat{j} = +29.4\hat{j}\ m/s$

$v_f = \frac{\vec{u}_1 + \vec{u}_2}{2} = 4.9\,m/s$

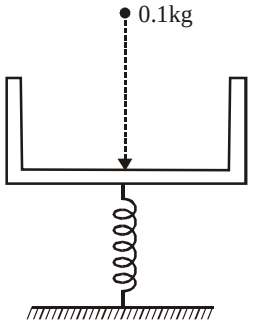
$h = 4.9 \times 2 - \frac{1}{2} \times 9.8 \times 2 \times 2 = 78.4m$

For the combined mass $x = ut + \frac{1}{2} \times 9.8\, t^2$, $t = 4.53$

$\therefore$ Total time of height $= 2 + 4.53 = 6.53$ sec.



18. A massless platform is kept on a light elastic spring, shown in Fig. When a sand particle of 0.1 kg mass is dropped on the pan from a height 0.24 m, the particle strikes the pan, and the spring compresses by 0.01 m. From what height (in meter) should the particle be dropped to cause a compression of 0.04 m ?



Ans. 3.96

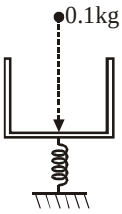
Sol. COME $\Rightarrow$ $mg(h + x_0) = \frac{1}{2}kx_0^2$

$\Rightarrow$ $mg(0.24 + 0.01)$

$= \frac{1}{2}k(0.01 \times 0.01)$ ....(i)

& $mg(h + 0.04) = \frac{1}{2}k(0.04)^2$ .....(ii)

Equation (i) divided by (ii) $h = 3.96$ m



PART-2 : CHEMISTRY

SECTION–I : (Maximum Marks : 18)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –1 In all other cases

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1. A mixture containing 0.05 mole of $K_2Cr_2O_7$ and 0.02 mole of $KMnO_4$ was treated with excess of KI in acidic medium. The liberated iodine required 1.0 L of $Na_2S_2O_3$ solution for titration. Concentration of $Na_2S_2O_3$ solution was :

(A) 0.40 mol L^{-1} (B) 0.20 mol L^{-1}

(C) 0.25 mol L^{-1} (D) 0.30 mol L^{-1}

Ans. A

Sol. eq mass of $K_2C_2O_7 = \dfrac{\text{molar mass}}{6}$

eq mass of $KMnO_4 = \dfrac{\text{molar mass}}{5}$

eq of $Na_2S_2O_3$ = eq of $K_2Cr_2O_7$ + eq of $KMnO_4$

or $N \times 1 = 0.02 \times 5 + 0.05 \times 6$

$\Rightarrow N = 0.4$ or $M = 0.4$

2. The energy of separation of an electron in a hydrogen like atom in excited state is 3.4 eV. The de-Broglie wave length (in Å) associated with the electron is :

(Given radius of first orbit of H-atom is 0.53Å)

(A) 3.33 (B) 6.66

(C) 13.31 (D) None of these

Ans. B

Sol. $2\pi r_n = n\lambda \Rightarrow 2\pi \times 0.53\dfrac{n^2}{z} = n\lambda$

$\lambda = 2\pi \times 0.53 \times \dfrac{n}{z}$

$E_{wep} = 3.4 = 13.6\dfrac{z^2}{n^2} \Rightarrow \dfrac{n}{z} = 2$

$\lambda = 2\pi \times 0.53 \times 2 = 6.66Å$

3. For a real gas (mol mass = 60) if density at critical point is $0.80\ g/cm^3$ and its $T_c = \dfrac{4\times10^5}{821}K$, then van der Waal's constant a (in atm L^2mol^{-2}) is

(A) 0.3375 (B) 3.375

(C) 1.68 (D) 0.025

Ans. B

Sol. $V_c = \frac{60}{0.80} = 75 cm^3\ mol^{-1}$

$b = \frac{V_c}{3} = 25 cm^3 mol^{-1} = 0.025 L\ mol^{-1}$

$\therefore T_c = \frac{8a}{27Rb}$

$\frac{4\times10^5}{821} = \frac{8\times a}{27\times0.0821\times0.025}$

$\Rightarrow a = 3.375$

4. What volume of 75% alcohol by weight $\left(d = 0.80 g/cm^3\right)$ must be used to prepare $150 cm^3$ of 30% alcohol by mass $\left(d = 0.90 g/cm^3\right)$?

(A) 67.5 mL (B) 56.25 mL

(C) 44.44 mL (D) 32.2 mL

Ans. A

Sol. Let V mL of alcohol be required

mass of alcohol is same in both solutions

$\therefore \frac{75}{100}\times0.8\times V = \frac{30}{100}\times0.9\times150$

V = 67.5 mL

5. Using MO theory predict which of the following species has the shortest bond length ?

(A) O_2^{2+} (B) O_2^{+}

(C) O_2^{-} (D) O_2^{2-}

Ans. A

Sol. Bond length $\propto \frac{1}{\text{Bond order}}$; B.O of $\underset{3}{O_2^{2+}}, \underset{2.5}{O_2^{+}}, \underset{1.5}{O_2^{-}}$ and $\underset{1}{O_2^{2-}}$

6. The bond dissociation energy of $B-F$ in BF_3 is $646 kJmol^{-1}$ where as that of $C-F$ in CF_4 is 515 kJ mol^{-1}. The correct reason for higher $B-F$ bond dissociation energy as compared to that of $C-F$ is :

(A) Smaller size of B atom as compared to that of C atom

(B) Stronger σ bond between B and F in BF_3 as compared to that between C and F in CF_4

(C) Significant $p\pi-p\pi$ interaction between B and F in BF_3 whereas there is no possibility of such interaction between C and F in CF_4

(D) Lower degree of $p\pi-p\pi$ interaction between B and F in BF_3 than that between C and F in CF_4

Ans. C

Sol. In BF_3 , B is sp^2 hybridized and has a vacant 2p orbital which overlaps laterally with a filled 2p orbital of F forming strong $p\pi-p\pi$ bond. However, in CF_4 the C does not have any vacant p-orbital.

SECTION– II : (Maximum Marks : 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

7. Which of the following statement is / are correct ?

(A) Tl^{3+} salts are oxidising agents (B) Ga^{+} salts are reducing agents

(C) Pb^{4+} salts are better oxidising agents (D) As^{5+} salts are oxidising agents

Ans. ABC

8. Which of the following oxides are amphoteric ?

(A) BeO (B) SnO

(C) ZnO (D) Al_2O_3

Ans. ABCD

9. Molecules with see-saw shape are :

(A) $XeOF_4$ (B) SF_4

(C) $XeOF_2$ (D) XeO_2F_2

Ans. BD

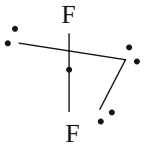
10. In which of the following species bond angle is expected to be more than 120° ?

(A) PCl_4^+ (B) NO_2^-

(C) NO_2^+ (D) XeF_2

Ans. CD

Sol. $O = \underset{sp}{\overset{+}{N}} = O$;



$Xe\left(sp^3d\right)$

11. Identify correct order of bond angles

(A) $Cl_2O > F_2O$ and $F_2O < H_2O$

(B) $AsI_3 > AsBr_3 > AsCl_3$

(C) $NO_2^+ > NO_2^-$

(D) $H_b\hat{B}H_b > H_t\hat{B}H_t$; where H_t is terminal Hydrogen of B_2H_6 and H_b is the bridging Hydrogen of B_2H_6

Ans. ABC

12. Which among the following compounds are paramagnetic ?

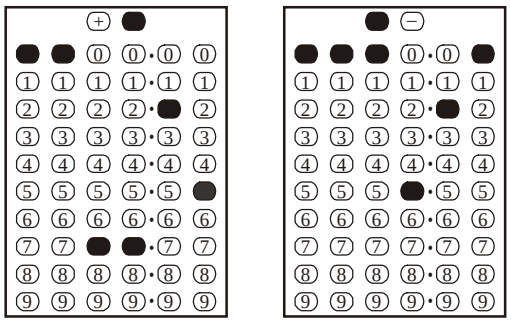
(A) KO_2 (B) K_2O_2

(C) K_2O (D) NO_2

Ans. AD

SECTION-III : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.
  For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +4 If ONLY the correct numerical value is entered as answer.
  Zero Marks : 0 In all other cases.

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13. 2 mole, equimolar mixture of $Na_2C_2O_4$ and $H_2C_2O_4$ required V_1L of 0.1 M $KMnO_4$ in acidic medium for complete oxidation. The same amount of the mixture required V_2L of 0.2 M NaOH for neutralization.Then the value of $(V_1 + V_2)$ is

Ans. 18.00

Sol. eq of $Na_2C_2O_4$ + eq of $H_2C_2O_4$

$= $ eq of $KMnO_4 = V_1 \times 0.1 \times 5$

$2 + 2 = 0.5\ V \quad ; \quad V_1 = 8L$

eq of $H_2C_2O_4$ = eq.of NaOH

$\Rightarrow 1 \times 2 = 0.2 \times V_2 \ ; \ V_2 = 10L$

14. If I excitation energy for the H-like (hypothetical) sample is 24 eV, then binding energy in III excited state in
eV is :

Ans. 12.00

Sol. $$\frac{\Delta E}{E_{4th}} = \frac{24}{E_{4th}} = \frac{\left[1-\frac{1}{4}\right]}{\left[\frac{1}{16}\right]} = \frac{3/4}{1/16}$$

$E_{4th} = 12$ eV

15. The van der Waal's constant 'b' of a gas is $4\pi \times 10^{-4}$ L / mol the centres of the two molecules approach each other is 10^{-x} m . What is x ? $N_A = 6 \times 10^{23}$

Ans. 10

Sol. $b = 4 \times \frac{4}{3} \times \pi r^3 \times N_A$

$4 \times \pi \times 10^{-4} \times 1000$

$= 4 \times \frac{4}{3} \times \pi \times r^3 \times 6 \times 10^{23}$

$r = 5 \times 10^{-9}\,cm$

Distance of closest approach = 2r

$= 10^{-8}\,cm$ or $10^{-10}\,m$

16. The wave function of atomic orbital of H like atoms is given as under

$$\psi_{2s} = \frac{1}{4\sqrt{2\pi}} z^{3/2} [2 - Zr] e^{Zr/2}$$

Given that the radius is in Å, then radius of nodal surface for He^+ ion is (in Å)

Ans. 1.0

17. The first ionisation energy of Li is 5.41 eV and electron affinity of chlorine is 3.61 eV.

The ΔH in $kJmol^{-1}$ for the reaction :

$Li(g) + Cl(g) \rightarrow Li^+(g) + Cl^-(g)$ is 1.737×10^x kJ mol^{-1}

What is the value of x ?

Ans. 2

Sol. $\Delta H = I.P \text{ of } Li + Eg \text{ of } Cl = 5.41 - 3.61 = 1.8eV$

$\Delta H\ mol^{-1} = 1.8 \times 6.023 \times 10^{23} eV$

$= 1.8 \times 6.023 \times 10^{23} \times 1.602 \times 10^{-19} \times 10^{-3} kJ$

$= 1.737 \times 10^2 kJ$

18. Among the triatomic molecules / ions , $BeCl_2, N_3^-, N_2O, NO_2^+, O_3, SCl_2, ICl_2^-, I_3^-$ and XeF_2, the total number of linear molecule (s) / ion(s) where the hybridization of the central atom does not have contribution from the

d-orbital(s) is :

Ans. 4

Sol. $BeCl_2, N_3^-, N_2O, NO_2^+$

PART-3 : MATHEMATICS

SECTION–I : (Maximum Marks : 18)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- Answer to each question will be evaluated according to the following marking scheme :

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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –1 In all other cases

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1. If terms independent of x in the expansion of $\left(3x-\frac{1}{x}\right)^{20}$ and $\left(x+\frac{\sqrt[9]{3^{10}}}{x}\right)^{18}$ are A and B respectively then $\left(\frac{9}{38}A+B\right)$ equals

(A) $3^{10}\cdot{}^{19}C_8$ (B) $3^{10}\cdot{}^{19}C_9$

(C) $3^{9}\cdot{}^{20}C_8$ (D) $3^{9}\cdot{}^{19}C_{10}$

Ans. B

Sol. Term independent of x in expansion of $\left(3x-\frac{1}{x}\right)^{20}$

$$T_{r+1}={}^{20}C_r\left(3x\right)^{20-r}\left(\frac{-1}{x}\right)^r$$

When r = 10

$$A=T_{11}={}^{20}C_{10}\,3^{10} \qquad \text{... (1)}$$

Term independent of x in expansion of $\left(x+\frac{\sqrt[9]{3^{10}}}{x}\right)^{18}$

$$T_{r+1}={}^{18}C_r\left(x\right)^{18-r}\left(\frac{\left(3^{10}\right)^{1/9}}{x}\right)^r$$

Where r = 9

$$B=T_{10}={}^{18}C_9\,3^{10} \qquad \text{... (2)}$$

So, $\left(\frac{9}{38}A+B\right)=\left(\frac{9}{38}\times{}^{20}C_{10}\times3^{10}+{}^{18}C_9\times3^{10}\right)$

[From eq. (1) and (2)]

$$=\left(\frac{9}{38}\times\frac{20}{10}\times\frac{19}{9}\times{}^{18}C_8\times3^{10}+{}^{18}C_9\times3^{10}\right)=3^{10}\times{}^{19}C_9$$

2. There are 3 women participating in a chess tournament. Every participant played 3 games with other participants. Number of games that the men played among themselves proved to exceed by 27 number of games that men played with women. Then number of men playing in the tournament is _____

(A) 3 (B) 5

(C) 9 (D) 12

Ans. C

Sol. Let number of men be n

Total number of games men played with women $=\left({}^{n}C_{1}\times{}^{3}C_{1}\right)3=3(3n)$

Now $=3\left({}^{n}C_{2}\right)-3(3n)=27$

$$\frac{n(n-1)}{2}-3n=9\Rightarrow n^{2}-7n=18\Rightarrow n^{2}-7n-18=0$$

$$\Rightarrow(n-9)(n+2)=0\Rightarrow n=9$$

3. The least integer 'a' for which $1+\log_{3}\left(x^{2}+1\right)\le\log_{3}\left(ax^{2}+4x+a\right)$ is true for all real values of x is

(A) 3 (B) 4

(C) 5 (D) 6

Ans. C

Sol. For $\log_{3}\left(ax^{2}+4x+a\right)$ to be valid

$ax^{2}+4x+a>0\ \forall\ x\in R$

$a>0$ and $4-a^{2}<0\Rightarrow a>2$

Given inequality $\log_{3}\left(\frac{ax^{2}+4x+a}{x^{2}+1}\right)\ge1$

$ax^{2}+4x+a\ge3\left(x^{2}+1\right)$

$(a-3)x^{2}+4x+a-3\ge0\ \forall\ x\in R$

For above expression to be true

$a>3$ and $\Delta\le0$

$16-4(a-3)^{2}\le0$

$a>3$ and $a-3\ge2$

So, $a\ge5$

4. The least value of x satisfying $\frac{1}{[x]}+\frac{1}{[2x]}=\{x\}+\frac{1}{3}$ is N, then number of positive divisors of 24N is

(where $[\cdot]$ is the greatest integer function and $\{x\}$ is fractional part of x)

(A) 2 (B) 4

(C) 6 (D) 8

Ans. B

Sol. Let $x = n + f$

$$\Rightarrow \frac{1}{n}+\frac{1}{2n+2f}=f+\frac{1}{3}$$

Case 1 : $0 \le f < \frac{1}{2}$

$$\Rightarrow \frac{1}{n}+\frac{1}{2n}=f+\frac{1}{3}$$

$$\Rightarrow \frac{1}{n}+\frac{1}{2n}=f+\frac{1}{3}$$

But $n=1 \Rightarrow f > \frac{1}{2}$

$\therefore\ n = 2, 3, 4$

Case 2 : $\frac{1}{2} \le f < 1$

$$\Rightarrow \frac{1}{n}+\frac{1}{2n+1}=f+\frac{1}{3} \ge \frac{1}{2}+\frac{1}{3}$$

$$\Rightarrow 10n^2-13n-16 \le 0$$

$\Rightarrow n = 1$ and $f = 1$

Not possible.

5. If $S_n(\theta)=\sum_{r=1}^{n}\frac{\cos(r-1)\theta}{(\cos\theta)^{r-1}}$, then sum of the series $\left[S_3\left(\frac{\pi}{3}\right)\right]+\left[S_9\left(\frac{\pi}{3}\right)\right]+\left[S_{15}\left(\frac{\pi}{3}\right)\right]+\ldots+\left[S_{2025}\left(\frac{\pi}{3}\right)\right]$ is

(where $[\cdot]$ denotes the greatest integer function)

(A) 0 (B) 1

(C) 3 (D) 2025

Ans. A

Sol. $S_n(\theta)=\text{Re}\left(\sum_{r=1}^{n}\frac{Z^{r-1}}{(\cos\theta)^{r-1}}\right)$ (where, $Z=\cos\theta+i\sin\theta$)

$$\text{Re}\left(\frac{1-\left(\frac{Z}{\cos\theta}\right)^n}{1-\frac{Z}{\cos\theta}}\right)=\text{Re}\left(\frac{1-\frac{\cos n\theta+i\sin^n\theta}{\cos^n\theta}}{-i\tan\theta}\right)$$

$$=\frac{\sin n\theta}{\sin\theta\cos^{n-1}\theta}$$

So, $S_{3\lambda}\left(\frac{\pi}{3}\right)=0$

6. If $2\cos^4 x+\sqrt{5}\left(\sqrt{5}-1\right)=\cos^2 x\left(\sqrt{5}+1+2\sin x\right)+\left(\sqrt{5}-1\right)\sin x$, then

$\sin^{10} x+4\sin^9 x+6\sin^8 x+4\sin^7 x+\sin^6 x+\sin x+1$ equals

(A) 1 (B) 2

(C) 3 (D) 4

Ans. B

Sol. $2\cos^4 x+\left(5-\sqrt{5}\right)=\cos^2 x\left(\sqrt{5}+1+2\sin x\right)+\left(\sqrt{5}-1\right)\sin x$

$\Rightarrow 2\cos^4 x-2\cos^2 x-\cos^2 x\left(2\sin x+\sqrt{5}-1\right)+2\left(2+\left(3-\sqrt{5}\right)\right)-\left(\sqrt{5}-1\right)\sin x=0$

$\Rightarrow 2\cos^4 x+2\sin^2 x-\cos^2 x\left(2\sin x+\sqrt{5}-1\right)+\left(3-\sqrt{5}\right)-\left(\sqrt{5}-1\right)\sin x=0$

$\Rightarrow \cos^4 x+\sin^2 x+\frac{3-\sqrt{5}}{2}-\cos^2 x\left(\sin x+\frac{\sqrt{5}-1}{2}\right)-\frac{\left(\sqrt{5}-1\right)}{2}\sin x=0$

$\Rightarrow \left(\cos^2 x\right)^2+\left(\sin x\right)^2+\left(\frac{\sqrt{5}-1}{2}\right)^2-\left(\sin x\right)\left(\cos^2 x\right)-\frac{\left(\sqrt{5}-1\right)}{2}\cos^2 x-\left(\frac{\sqrt{5}-1}{2}\right)\sin x=0$

$\Rightarrow \left(\cos^2 x-\sin x\right)^2+\left(\sin x-\frac{\sqrt{5}-1}{2}\right)^2+\left(\cos^2 x-\frac{\sqrt{5}-1}{2}\right)^2=0$

$\Rightarrow \cos^2 x=\sin x=\frac{\sqrt{5}-1}{2}$

$\Rightarrow \cos^2 x=\sin x$

$\Rightarrow \sin^2 x+\sin x=1$ ... (i)

$\sin^6 x+4\sin^5 x+6\sin^4 x+4\sin^3 x+\sin^2 x+\sin x+1$

$=\sin^2 x\left(\sin^2 x+\sin x\right)^4+\sin x+1$ from (i)

$=\sin^2 x+\sin x+1=2$

SECTION– II : (Maximum Marks : 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

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7. In a game a person P at (x, y) can jump to any one of the following points relatively (x + 2, y), (x + 4, y), (x, y + 2), (x, y + 4). If P is at (1, 1) then the number of ways to reach the point.

(A) (5, 5) is 14 (B) (5, 5) is 24

(C) (7, 8) is 0 (D) (9, 9) through (5, 5) is 196

Ans. ACD

Sol. A = (5, 5), B = (9, 9), C = (7, 8)

Let $L_1 = (x+2,\ y) \rightarrow$ Horizontally 2 units

$L_2 = (x+4,\ y) \rightarrow$ Horizontally 4 units

$U_1 = (x,\ y+2) \rightarrow$ Vertically 2 units

$U_2 = (x,\ y+4) \rightarrow$ Vertically 4 units

To reach from P to A 4 units horizontally and 4 units vertically to be travel

$$L_1L_1U_1U_1 = \frac{4!}{2!\,2!} = 6 \text{ ways}$$

$$L_1L_1U_2 = \frac{3!}{2!} = 3 \text{ ways}$$

$$L_2U_1U_1 = \frac{3!}{2!} = 3$$

$$L_2U_2 = 2! = 2$$

Total number of ways = 14

Number of ways to travel from P to B

= (number of ways from P to A). (number of ways from A to B)

$= 14 \times 14 = 196$

8. If the equation $(2+\sin^2 x)^2 - (a-3)(2+\sin^2 x)(1+\sin^2 x) + (a-4)(1+\sin^2 x)^2 = 0$ has atleast one real root then the complete set of values of a is $[\alpha,\ \beta]$ then

(A) $\alpha + \beta = 11$ (B) $\beta - \alpha = \frac{1}{2}$

(C) $\alpha+\beta=\frac{33}{2}$ (D) $\beta-\alpha=1$

Ans. BC

Sol. Let $\frac{2+\sin^2 x}{1+\sin^2 x}=t\Rightarrow t\in\left[\frac{3}{2},\ 2\right]$

$t^2-(a-3)t+(a-4)=0$

$\Rightarrow t=1$ or $a-4$

Clearly $a-4\in\left[\frac{3}{2},\ 2\right]$

$a\in\left[\frac{11}{2},\ 6\right]$

9. Let $S=\sum_{k=1001}^{3001}\left(\frac{1}{k}\right)$, then

(A) $S>1$ (B) $S<1$

(C) $S>\frac{4}{3}$ (D) $S<\frac{4}{3}$

Ans. AD

Sol. AM > GM

$\Rightarrow \sum k\sum\frac{1}{k}>(2001)^2$

$\Rightarrow \sum\frac{1}{k}>1\Rightarrow S>1$

Also

$S<\frac{500}{1000}+\frac{500}{1500}+\frac{500}{2000}+\frac{500}{2500}+\frac{1}{3001}$

$S<\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\frac{1}{3000}=\frac{3851}{3000}<\frac{4}{3}$

10. Let $N=2^2\times3^3\times7^4$, then

(A) Number of factors of N^2 which are less than N and are not the factors of N is 98

(B) Number of factors of N which are of the form 4K + 1, $K\in N$ is 9

(C) Number of factors of N which are multiples of 54 is 10

(D) Number of factors of N which are of the form 4K + 1, $K\in N$ is 8

Ans. ABC

Sol. $N=2^4\times3^6\times7^8$

(A) Total number of factor $=5\times7\times9=315$

Number of factors less than N = 156

Number of factors of N = 58

$\therefore$ Answer = 156 – 58 = 98

(B) If the factor of the form 3^{α}, 7^{β}, where $\alpha+\beta$ is

Even is of the form 4K + 1

Answer = 9

(C) $\frac{N}{54}=2\times7^4$

Number of factors = 10

11. The series $\frac{8}{5}+\frac{16}{65}+\frac{24}{325}+.....\frac{8n}{4n^4+1}$

(A) Sum of infinite number of terms of series is 4

(B) Sum of n terms of series is less than 2 $\forall\ n\in N$

(C) Sum to n terms of series cannot be an integer for any $n\in N$

(D) Sum of infinite number of terms of series is less than 3

Ans. BCD

Sol. r^{th} term $T_r=\frac{8r}{4r^4+1}$

$$=\frac{8r}{\left(2r^2+2r+1\right)\left(2r^2-2r+1\right)}$$

$$=2\left\{\frac{2r^2+2r-1-\left(2r^2-2r+1\right)}{\left(2r^2+3r+1\right)\left(2r^2-3r+1\right)}\right\}$$

$$=2\left\{\frac{1}{2r^2-2r+1}-\frac{1}{2r^2+2r+1}\right\}$$

$$=2\left\{\frac{1}{v_r}-\frac{1}{v_{r+1}}\right\}$$

$$\sum_{r=1}^{\infty}T_r=2\left[\left(\frac{1}{1}-\frac{1}{5}\right)-\left(\frac{1}{5}-\frac{1}{13}\right)+.....\right]=2$$

12. Consider 5 points in plane are situated so that no two of the straight lines joining them are parallel, perpendicular, or co incident. From each point perpendiculars the drawn to all the lines joining the other four points. If k is the maximum number of intersections that these perpendiculars can have ? Then

(A) More than 310 (B) k is divisible by 5

(C) k is less than 400 (D) k is even number

Ans. ABC

Sol. Lets consider 5 points A, B, C, D and E. Consider a point A, number of straight lines that can be drawn through B, C, D and E is $\left({}^4C_2\right)=6$ straight lines, so from A we can draw perpendicular to these 6 straight lines, similarly we will get 6 perpendiculars form other points as well so total number of perpendiculars is $6\times5=30$. The maximum number of intersections of these 30 straight lines is $\left({}^{30}C_2\right)=435$, but these 435 points are not distinct means out of 30 straight lines not all of them are non concurrent. We have to consider the following cases :

Case (I) : Consider one of the ten lines drawn from original 5 points, say this line AB, there are three perpendiculars are drawn on AB from points C, D and E, these 3 perpendiculars are parallel to each other hence they will not intersect each other, these three point if not parallel could have intersected at 3 points hence we lost $3\times10=30$ points.

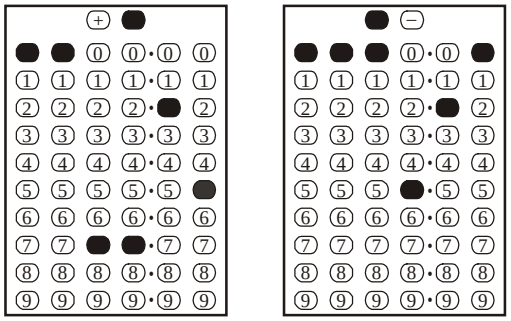
Case (II) : Since three altitudes of a triangle intersect each other at a point (called ortho center). From the original 5 points we will have $\left({}^{10}C_3\right)=10$ triangles. Consider one of these 10 triangles, perpendicular from vertex to opposite side (altitude) will intersect at one point instead of 3 points, hence from each triangle we lost 2 points so total number of points that we have to subtract is $2\times10=20$ points.

Case (III) : Consider one of the original 5 points (say A) since form A we have drawn 6 perpendiculars and these 6 perpendiculars are concurrent hence they intersect each other at one point instead of $\left({}^{6}C_2=15\right)$ points so from one of this point we lost $15-1=14$ points, hence total we lost $14\times5=70$ points.

So total number of points is $435-30-20-70=315$ points.

SECTION-III : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.
  For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +4 If ONLY the correct numerical value is entered as answer.
  Zero Marks : 0 In all other cases.

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13. If $S=\lim\limits_{n\to\infty}\sum\limits_{r=2}^{n}\frac{{}^{r}C_2}{3^r}$. Then value of 48 S is

Ans. 18

Sol. $S=\lim\limits_{n\to\infty}\sum\limits_{r=2}^{n}\frac{{}^{r}C_2}{3^r}$

$$=\lim_{n\to\infty}\left(\frac{1}{9}+\frac{3}{27}+\frac{6}{81}+....\right)$$

$$S=\frac{1}{9}+\frac{3}{27}+\frac{6}{81}+....\infty$$

$$\frac{S}{3}=\frac{1}{27}+\frac{3}{81}+....\infty$$

$$\frac{2S}{3}=\frac{1}{9}+\frac{2}{27}+\frac{3}{81}....\infty$$

$$\frac{2S}{9}=\frac{1}{27}+\frac{2}{81}+....\infty$$

$$\frac{2S}{3}-\frac{2S}{9}=\frac{1}{9}+\frac{1}{27}+\frac{1}{81}+....\infty$$

$$\frac{4S}{9}=\frac{\frac{1}{9}}{1-\frac{1}{3}}$$

$$\Rightarrow\frac{4S}{9}=\frac{1}{6}$$

$$\Rightarrow S=\frac{3}{8}$$

$$\Rightarrow S=0.38$$

14. If α, β are two roots of the equation $1!+2!+3!+.....(x-1)!+x!=k^2$, $k\in I^+$. Where $\alpha<\beta$ and if α_1, α_2, α_3, α_4 are roots of equation $\left(a+\sqrt{b}\right)^{x^2-\left(\alpha+3\alpha+5\alpha^2\right)}+\left(a-\sqrt{b}\right)^{x^2-3\beta}=2a$, where $a^2-b=1$.

Then the value of $|\alpha_1+\alpha_2+\alpha_2+\alpha_3+\alpha_1\alpha_2\alpha_3\alpha_4|$ is equal to

Ans. 80

Sol. 1! + 2! + 3! + 4! = 33

For $x\geq 4$ unit place of $\sum_{r=1}^{x} r'$ is 3

But square of any number cannot have unit place 3

$\Rightarrow x<4$ is only possibility

for x = 1 $\quad k^2=1\Rightarrow k=1$

for x = 3 $\quad k^2=9\Rightarrow k=3$

$\Rightarrow \alpha=1,\ \beta=3$

Now, $\left(a+\sqrt{b}\right)^{x^2-(1+3+5)}+\left(a-\sqrt{b}\right)^{x^2-3\times3}=2a$

$$\left(a+\sqrt{b}\right)^{x^2-9}+\left(a-\sqrt{b}\right)^{x^2-9}=2a$$

As $a^2-b=1$, then only possibility of above equation is when

$x^2-9=1$ or $x^2-9=-1$

$x^2=10$ or $x^2=8$

Hence, $x \pm \sqrt{10},\ \pm\sqrt{8}$

$\left|\alpha_1+\alpha_2+\alpha_3+\alpha_4+\alpha_1\alpha_2\alpha_3\alpha_4\right|$

$=\left|+\sqrt{10}-\sqrt{10}+\sqrt{8}-\sqrt{8}+\left(\sqrt{10}\right)\left(\sqrt{-10}\right)\left(\sqrt{8}\right)\left(\sqrt{-8}\right)\right|=80$

15. The real value of x satisfying $\sqrt[3]{20x+\sqrt[3]{20x+13}}=13$ can be expressed as $\frac{a}{b}$ where a and b are relatively prime positive integers. Then the value of $(a-100b)$ is

Ans. 46

Sol. Consider the pattern

$$a=\sqrt[3]{20x+\sqrt[3]{20x+\sqrt[3]{20x+....\sqrt[3]{20x+\sqrt[3]{20x+13}}}}}$$

$$a=\sqrt[3]{20x+\sqrt[3]{20x+\sqrt[3]{20x+....\sqrt[3]{20x+13}}}}$$

No change (infinitely many terms)

Hence, $\sqrt[3]{20x+13}=13$

$$x=\frac{13^3-13}{20}=\frac{546}{5}$$

16. Number of order pairs (x, y) which satisfy the equation $x^2+2x\sin(xy)+1=0,$ where $y\in[0,\ 2\pi]$ is

Ans. 02

Sol. Given equation is $x^2+2x\sin(xy)+1=0$

$\Rightarrow x^2+2x\sin(xy)+\sin^2(xy)+1-\sin^2(xy)=0$

$[x+\sin(xy)]^2+\cos^2(xy)=0$

$\sin(xy)=-x$ ... (i) and $xy=\pm\frac{\pi}{2},\ \pm\frac{3\pi}{2}$

Let $xy=\frac{\pi}{2}$

(i) $\Rightarrow 1=-x\Rightarrow x=-1$

$(-1)y=\frac{\pi}{2}\Rightarrow y=-\frac{\pi}{2}\notin[0,\ 2\pi]$

Let $xy=-\frac{\pi}{2}$

(i) $\Rightarrow -1=-x\Rightarrow x=1$

(i) $=-\frac{\pi}{2}=y=-\frac{\pi}{2}\notin[0,\ 2\pi]$

Let $xy=\frac{3\pi}{2}$

(i) $\Rightarrow -1=-x\Rightarrow x=1$

(i) $y=\frac{3\pi}{2}\Rightarrow y=\frac{3\pi}{2}\in[0,\ 2\pi]$

Let $xy = -\frac{3\pi}{2}$

(i) $\Rightarrow 1 = -x \Rightarrow x = -1$

$(-1)y = -\frac{3\pi}{2} \Rightarrow y = \frac{3\pi}{2} \in [0,\ 2\pi]$

$(x,\ y) = \left(1, \frac{3\pi}{2}\right), \left(-1,\ +\frac{3\pi}{2}\right)$

Numbers of order pairs = 2.

17. Let $P(x) = x^6 - x^5 - x^3 - x^2 - x$ and $\alpha,\ \beta,\ \gamma,\ \delta$ are the roots of the equation $x^4 - x^3 - x^2 - 1 = 0$, then $P(\alpha) + P(\beta) + P(\gamma) + P(\delta) =$

Ans. 06

Sol. Let $Q(x) = x^4 - x^3 - x^2 - 1 = 0$ has roots $\alpha,\ \beta,\ \gamma,\ \delta$ and

$P(x) = (x^6 - x^5 - x^4 - x^2) + (x^4 - x^3 - x^2 - 1) + (x^2 - x + 1)$

$\Rightarrow P(x) = (x^2 + 1)Q(x) + x^2 - x + 1$

$\sum P(\alpha) = \sum \alpha^2 - \sum \alpha + 4$

$= \left(\sum \alpha\right)^2 - 2\sum \alpha\beta - \sum \alpha + 4$

$= (1)^2 - 2(-1) - (1) + 4 = 6$

18. The result of 22 hockey matches are to be predicted. Given that result can be win, lose or draw. If p is number of different forecast which contain exactly 18 correct results and q is number of different forecast which contains exactly 19 correct results. Then value of $\left(\frac{4p}{q}\right)$ is

Ans. 38

Sol. $p = {}^{22}C_{18} \times 2 \times 2 \times 2 \times 2$

$q = {}^{22}C_{19} \times 2 \times 2 \times 2$

$\frac{p}{q} = \frac{\frac{22 \times 21 \times 20 \times 19}{4 \times 3 \times 2 \times 1}}{\frac{22 \times 21 \times 20}{3 \times 2}} \times 2 = \frac{19}{2} = 9.5$